

Ian Zhang

Curriculum Vitae

Contact Information

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Google Scholar: <https://scholar.google.ca/citations?user=0FL5fjMAAAAJ&hl=en>

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Education

PhD in Statistics, University of Toronto

Sep 2026 -

- Advisors: Joshua Speagle, Thibault Randrianarisoa

MSc in Statistics, University of Toronto

Sep 2025 - Jun 2026

- Advisor: Nancy Reid

Honours BSc in Statistics/Mathematics, University of Toronto

Sep 2021 - Jun 2025

- Advisor: Jun Young Park
- Graduated with high distinction

Publications and Preprints

[†]: Corresponding Author;

*: Equal contribution

Published

1. Michael A. Kouritzin, **Ian Zhang**[†], Jyoti Bhadana, Seoyeon Park. Markov processes for enhanced deepfake generation and detection[J]. *AIMS Mathematics*, 2026, 11(4): 11731-11759. doi: 10.3934/math.2026483

Under Review

1. Aditya Khan, **Ian Zhang**

2026

Parametric Hotspot Analysis with the Getis-Ord Statistic

In Progress

1. **Ian Zhang**[†], Thibault Randrianarisoa

2026

Likelihood Tempering to Mitigate Prior Dominance in Variational Posteriors for Bayesian Neural Networks

2. **Ian Zhang**^{*,†}, Hanlong Chen^{*}, . . . , Howard Chertkow, . . . , Malcolm Binns *2026*
Redundancy-Aware Model Assessment for Component and Cluster Selection in Clinical Subtype Discovery
3. Hanlong Chen^{*}, **Ian Zhang**^{*}, . . . , Malcolm Binns *2026*
Continuous Cognitive Heterogeneity in Dementia: A Severity Factor and an Age-Related Amnesic Axis Without Reproducible Subtypes
4. **Ian Zhang**[†], Finn Tran, Meredith Franklin *2026*
Spatial Dependence Regularization for Improved Generalization in Climate Downscaling Models
5. Gabriel Liu, **Ian Zhang**^{*,†}, . . . , Daniel Alessi *2026*
LiBRE: Statistical Learning for Representative Brine Design and Lithium Enrichment Analysis

Research Experience

- Doctoral Student**, Department of Statistical Sciences/University of Toronto *2026-Present*
- Graduate Research Assistant**, Baycrest Institute/University of Toronto *2025-Present*
Supervisor: Malcolm Binns
- Summer Research Trainee**, McGill University *2025*
Supervisor: Jianguo (Jeff) Xia
- Research Assistant**, University of Alberta *2023-2024*
Supervisor: Mike Kouritzin

Teaching

Teaching Assistant

2025-2026 School Year

- STA302: Methods of Data Analysis I** *Fall 2025*
- MAT137: Calculus with Proofs** *Fall 2025 & Winter 2026*
- STA437: Methods of Multivariate Data** *Winter 2026*
- UofT Statistical Sciences Research Program (UTSSRP)** *Summer 2026*

2026-2027 School Year

- MAT148: Calculus I with Proofs** *Fall 2026*
- STA257: Probability and Statistics I** *Fall 2026*
- MAT149: Calculus II with Proofs** *Winter 2027*
- STA457: Time Series Analysis** *Winter 2027*

Presentations

Conference Talks

2026 SSC Annual Meeting in Hamilton 2026

Presented *Likelihood Tempering to Mitigate Prior Dominance in Variational Posteriors for Bayesian Neural Networks* in the New Frontiers in Bayesian Analysis session.

2025 SSC Annual Meeting in Saskatoon 2025

Presented *Markov Processes for Enhanced Deepfake Generation and Detection* in the Probability session.

Third Joint SIAM/CAIMS Annual Meetings 2025

Presented *Markov Processes for Enhanced Deepfake Generation and Detection* in the Machine Learning and Stochastic Processes session.

Posters

The Thirteenth Annual Canadian Statistics Student Conference 2025

Presented *Comparing Exact and Approximate Inference for Spatial Autocorrelation Tests: A Power Analysis*.

Honours and Awards

Anna & Alex Beverly Memorial Fellowship 2025

Awarded by University College (University of Toronto) for general proficiency during my undergraduate degree.

NSERC Undergraduate Student Research Award (x2) 2023, 2024

Awarded by the University of Alberta to research Markov process methods for deepfake generation and detection.

Jack Zwaigenbaum Scholarship 2022

Awarded by University College (University of Toronto) for academic excellence in my first year of undergraduate studies.

Dean's List Scholar (x4) 2022, 2023, 2024, 2025

Awarded by the Faculty of Arts and Science (University of Toronto) for maintaining academic excellence every year of my undergraduate degree.

Grants

CSSC Travel Grant 2025

Awarded to full-time Canadian students presenting a poster or giving a talk at the Canadian Statistics Student Conference.

Services

Conferences

Session chair, 2025 Third Joint SIAM/CAIMS Annual Meetings

Chaired the “Machine Learning and Stochastic Processes” session.

Other

Citizenship: Canadian

Languages Spoken: English (native), Mandarin (native), Japanese (intermediate)

Memberships: Statistical Society of Canada